

Supporting or Hindering? Understanding the Divergent Effects of LLM Assistance on Critical Thinking in Academic Reading

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Abstract

Student researchers increasingly use large language models (LLMs) to support academic reading, yet their effects on critical thinking remain unclear. We hypothesize that research experience moderates the impact of LLM assistance on student researchers' critical thinking. We conducted a user study with 13 student HCI researchers (5 experienced, 8 novice), asking them to read academic papers with and without LLM assistance and write critical reviews. We find divergent effects: LLM assistance enhances critical thinking among experienced researchers, but tends to diminish it among novices, with substantial individual variability. Based on these findings, we discuss design implications for maintaining human agency while delegating cognitive labor to LLMs in academic work.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Large Language Models, LLMs, LLM Assistant, Human-AI Collaboration, Academic Reading, Paper Understanding, Reading Comprehension, Critical Thinking, Expertise, Researchers, AI for Science

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1 Introduction

Reading research papers with large language models (LLMs) is becoming common among student researchers [1, 11, 15]. Yet, it remains unclear how LLMs affect the reading experience and the development of critical thinking skills. For example, some scholars warn that students may delegate key cognitive tasks to LLMs and thereby lose opportunities to practice core competencies, such as critically evaluating papers and constructing arguments [4], but whether these concerns hold in practice is still unknown. This raises questions about whether widely used LLM-assisted reading tools effectively support student researchers and, if not, how students should be guided to use LLMs in ways that foster critical thinking.

To address this gap, we test the hypothesis that research experience moderates how LLM assistance influences student researchers' critical thinking. We contribute a user study with 13 student HCI researchers (5 experienced and 8 novice, based on peer-review experience) that evaluates critical thinking during paper reading. Participants read papers with and without an LLM assistant and write a review of strengths, weaknesses, and suggestions for improvement. We then evaluate the degree to which participants engaged in critical thinking while reading a paper using the critical thinking rubric [2].

Our results show that LLM assistance has contrasting effects on critical thinking: experienced student researchers benefit from LLM support, whereas novices experience a decline in critical thinking performance. Our analysis of behavior logs suggests that this is because, compared to novices, experienced researchers spend more time engaging with the original papers and use the LLMs primarily as a supplementary aid. We further find heterogeneity in critical thinking performance and behavior pattern within the novice group: those who use LLMs as a supplemental assistant, similar to experienced students, tend to have improved performance, whereas those who reported relying on and trusting the LLMs tend to have degraded performance. Based on these findings, we discuss the design implications on how the systems that provide LLM support to student researchers can sustain their human agency [3, 14].

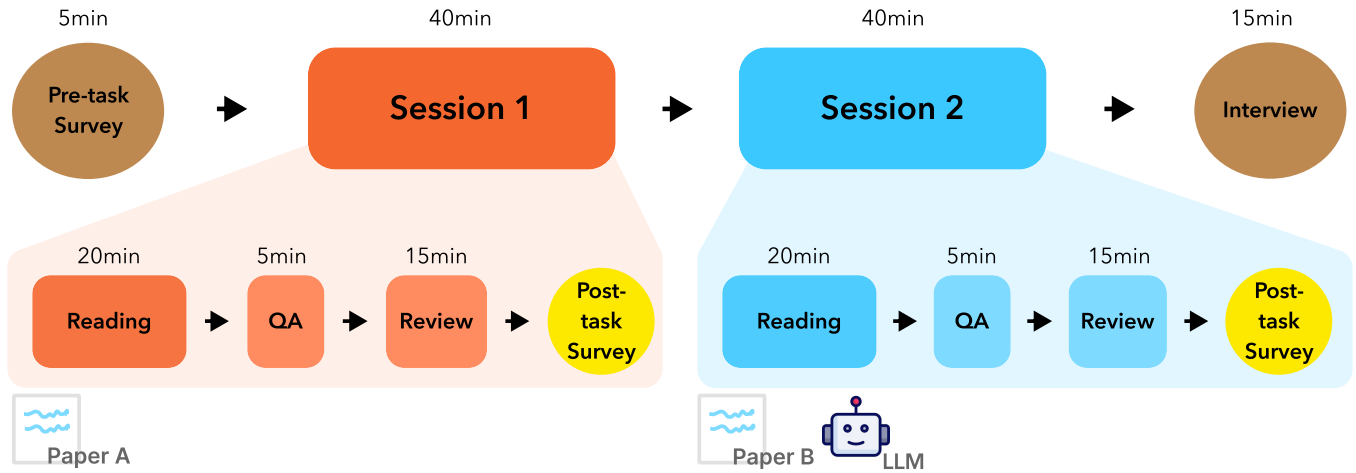


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* The order of conditions(+LLM/-LLM) and papers (A/B) is counterbalanced.

Figure 1: The procedure of our user study. We investigate how LLM assistance influences participants' critical thinking by guiding them to read academic papers with and without LLM support across two sessions.

2 User Study

We discuss our study design for investigating how LLM assistant shapes student researchers' critical thinking.

2.1 Objectives

We hypothesize that LLM assistance has divergent effects on student researchers' critical thinking, depending on their level of research experience. To test this hypothesis, we pose two research questions:

- (RQ1) How does LLM assistance affect the critical thinking of experienced and novice student researchers?
- (RQ2) What are the underlying causes of this difference?

We operationalize research experience by whether participants had prior peer-review experience for major conferences. We adopt this binary characterization because continuous proxies such as seniority do not necessarily reflect expertise. In contrast, being invited to serve as a reviewer provides a stronger signal of student researchers' competence. By answering these questions, we aim to obtain guidance on the design and use of LLM-assisted tools to support student researchers (Sect. 4).

2.2 Study Design

Our study consists of four phases (Fig. 1): (1) a pre-task survey, (2) the main experiment, (3) a post-task survey, and (4) a post-study interview. In the pre-task survey, participants report their experience with LLMs, academic background, and demographics. Then, participants move to the main experiment that comprises two sessions, each consisting of three rounds: reading (20 minutes), question-answering (5 minutes), and review writing (15 minutes). Each participant completed two conditions under different LLM availability (with or without), with the order of conditions and paper assignments counterbalanced across participants. We conduct post-task surveys after each session, and a semi-structured interview at the end to explore participants' reading strategies and experiences across conditions. The detailed study design is as follows:

Participants. We recruit 13 student HCI researchers through a university forum. Inclusion criteria are: (i) graduate students in HCI or adjacent fields, and (ii) frequent LLM users (more than three times per week) in everyday study or research. We recruit 5 novices and 8 experienced student researchers.

Study interface. We develop a web-based interface for the study (Fig. 2). The interface consists of a paper viewer on the left panel and an LLM chat window on the right panel (Fig. 2b), which is enabled only in the with-LLM condition. Participants can use the chat interface to freely interact with the paper using natural-language prompts, e.g., by asking questions or requesting summaries or clarifications of unfamiliar terms (Fig. 2a). We also enable participants to select specific texts and query related questions (Fig. 2c). We do not allow external web search to isolate the effect of LLM assistance. We use OpenAI gpt-5.1 for our LLM agent.

Measures. We assess the degree of critical thinking by the quality of participants' written reviews (RQ1), evaluated using a Critical Thinking VALUE Rubric [2]. This rubric assesses the level of critical thinking in five dimensions: explanation of issues, use of evidence, influence of context and assumptions, student's position, and conclusions. We use the automatic evaluation framework of Kim et al. [9] that rates the reviews based on a four-point scale. We also examine reading behaviors from system logs (RQ2), classifying reading patterns into three categories based on dwell time: focused reading (≥ 5 seconds), scanning (2–5 seconds), and scrolling (< 2 seconds). From these classifications, we compute the focused reading ratio (i.e., the proportion of time spent in focused reading) and the focused reading time.

3 Results and Discussion

We describe our study findings in both quantitative and qualitative perspectives.

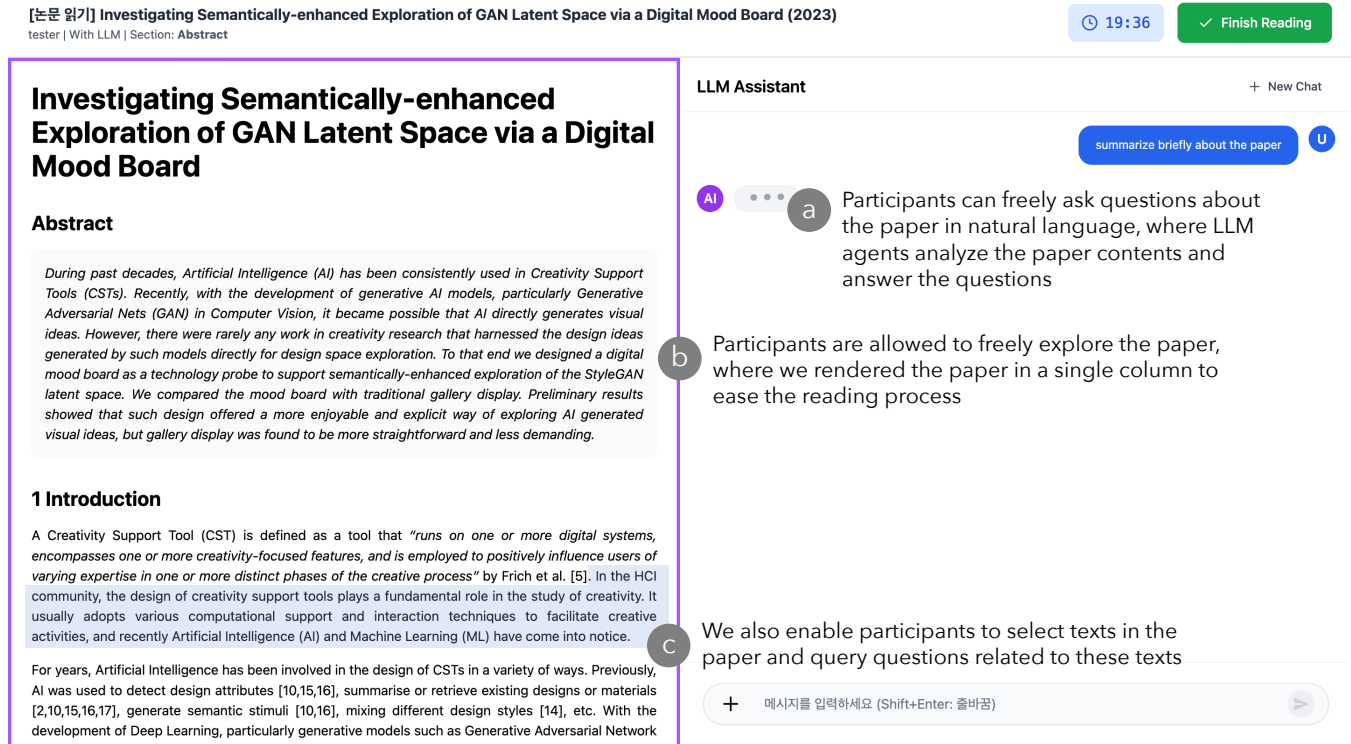


Figure 2: The user interface used for our experiment. Participants are allowed to freely read papers and generate questions, both about the entire document (a) and about the specific parts of the paper (b). The chat interface is disabled for without-LLM condition.

3.1 Quantitative Findings

We discuss our quantitative findings on the observed level of critical thinking and reading behavior.

The level of critical thinking. We verify that LLM provides a consistent benefit for experienced researchers while having mixed effects on novices. We find that experienced participants' review quality, a proxy for critical thinking, increases with LLM support on average ($\Delta = +0.20$; Fig. 3a). Meanwhile, the review quality of novices declines ($\Delta = -0.13$; Fig. 3b). Furthermore, novices exhibit higher individual variability ($SD = 0.61$) in review quality compared to experienced ($SD = 0.32$) (Fig. 3a). We also examine how participants' performance varies at the individual level. By visualizing changes in review quality between the with- and without-LLM conditions (Fig. 3b), we find that experienced participants consistently benefit from LLM support in critical thinking, whereas novices exhibit divergent results: five show declines, two show improvements, and one remains unchanged.

A closer examination of score changes across the sub-dimensions of the critical thinking rubric further specifies the contrasting effects of LLM. (Fig. 4) Among the five sub-dimensions, experienced participants showed improvements in explanation quality, use of evidence, and conclusions. Performance in the context dimension remained stable ($\Delta = 0.00$). However, position dimension demonstrates a larger decrement for experienced participants ($\Delta = -0.60$) than for novices ($\Delta = -0.25$). This suggests that LLMs

can effectively support critical reasoning for experienced readers, but may also be used to confirm and reinforce their prior perspectives. Novices exhibited the same trends as those of experts across sub-dimensions, except that their evidence evaluation scores declined with LLM use ($\Delta = -0.38$), whereas experts showed improvement ($\Delta = +0.40$). This suggests that when novices use LLMs, they may engage less with the original text, thereby weakening their ability to develop arguments grounded in specific details and evidence. This interpretation is consistent with our analysis of the focused reading ratio and time, presented below.

Focused reading ratio and time. We find that the focused reading ratio slightly decreased for the experienced (-5.2%) but dropped more substantially for Novices (-10.8%). This shift in reading strategy is closely associated with critical thinking outcomes within the novice group. Novices whose critical thinking improved showed reading patterns similar to experienced participants, exhibiting only a small decrease in reading ratio (-2.1%) and spending more time in focused reading (890 sec vs. 709 sec for those whose critical thinking declined). These results provide additional explanations on (1) why LLM assistance generally enhances critical thinking among experienced participants but has mixed effects for novices, and (2) why novices often fail to provide concrete evidence in their reviews.

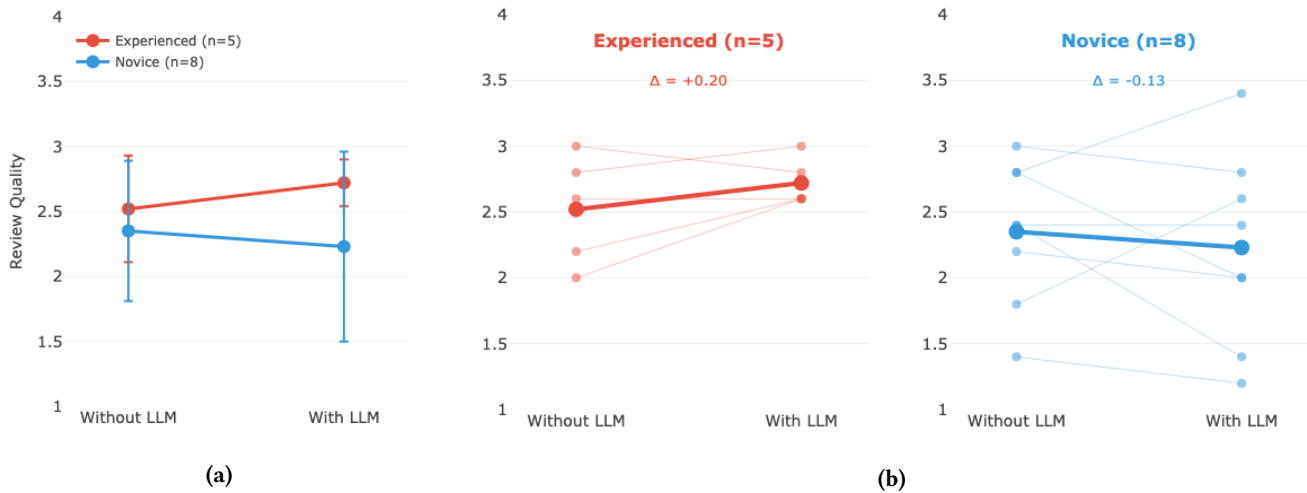


Figure 3: The change of review quality as a proxy for critical thinking by the existence of LLM support. (a) We find that experienced participants (red) improve their review quality when assisted by an LLM, whereas novices (blue) show a decline. (b) We further examine individual-level performance and observe that experienced participants consistently benefit from LLM support in critical thinking. In contrast, novices exhibit mixed effects: five show declines, two show improvements, and one remains unchanged.

3.2 Qualitative Findings from Interview

We discuss our qualitative findings obtained from the post-hoc interview. The findings align with the quantitative findings (Sect. 3.1) that experienced participants benefit from LLMs in enhancing their critical thinking, while novices often do not.

Experienced participants demonstrate skepticism on LLM output. Aligned with our quantitative findings, experienced participants frequently expressed skepticism toward LLM outputs. For example, P1 note: “I use LLMs all the time, but I don’t really trust them... I usually just get a general sense of things first, and then I verify them myself while reading.” Similarly, P2 remark: “I actually don’t tend to use AI when I’m seriously reading papers... when I read properly, I just read properly.” Meanwhile, experts also report the temptation to become complacent when using LLMs and note that they had to actively resist this tendency. This pattern suggests that, whereas novices may more readily fall into such complacency, experience plays an important role in mitigating it. These findings indicate potential danger in providing unconstrained LLM support to novices without additional scaffolding (Sect. 4).

Novices exhibit varying degrees of reliance on LLM. In contrast to experienced participants, novice researchers expressed divergent attitudes toward LLM outputs: some actively verify the outputs, whereas others rely heavily on the system. For example, P9, P10, and P12, express strong trust in and reliance on LLMs. P9 remark: “Because I thought the LLM would make things easier, I think I ended up reading more superficially... it also felt like my level of concentration was weaker than usual.” Meanwhile, P7 and P13 exhibit the same skepticism toward LLM outputs observed in the experienced group. P7 noted: “The abstract is already an accurate summary written by the authors, so I kept wondering why I should even ask an LLM to do that.” and added: “I tried to use the LLM only

as a supplementary tool.” P13 similarly state: “I couldn’t fully trust the LLM’s answers 100%.”

4 Discussion: Design Implications for AI Delegation

Our study provides design implications for the future system that leverages LLM to support student researchers. In summary, we recommend that such systems should dynamically adapt to students’ expertise, thereby maximizing the benefits of LLM support while preserving their agency [12, 14].

Provide interventions that guide novices to properly delegate tasks to LLMs.. We find that novice student researchers often over-rely on LLMs; for example, they draw on fewer pieces of evidence and provide weaker justifications (Sect. 3.1). This issue reflects a widely known AI-delegation challenge: novices may not know when or how to effectively delegate their labor to AI [5–8]. Although we observed overuse, underuse of AI also frequently occurs [13]. Accordingly, we argue that LLM-assisted systems should detect when novice student researchers are overusing or underusing the LLM and provide interventions that guide them toward appropriate delegation. Doing so likely requires revisiting traditional human-centered design approaches [10]: characterizing users’ task workflows and defining rules (or guidance policies) that specify when delegation is beneficial.

Introduce diverse perspectives to experts. We find that, for experienced researchers, LLMs can effectively support clearer and more comprehensive articulation of issues and more logically structured conclusions, but they may also be used to confirm and reinforce users’ prior perspectives, i.e., being biased. This observation aligns with prior findings that LLMs can amplify human bias in

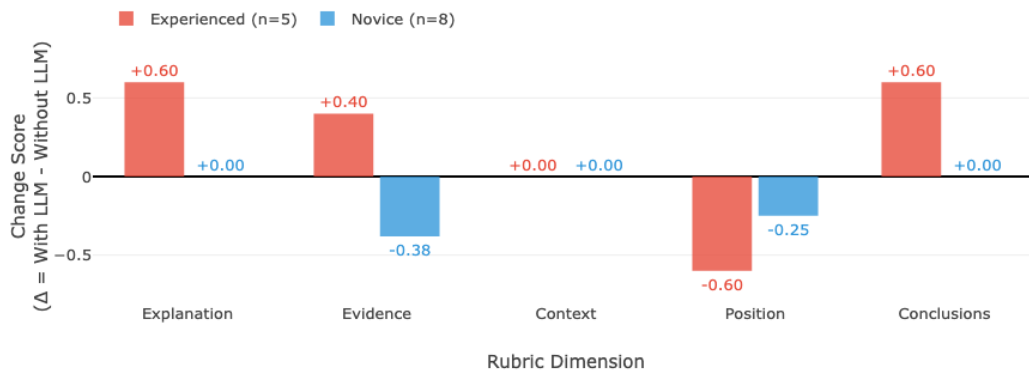


Figure 4: Changes in scores across the sub-dimensions of the critical thinking rubric. Experienced participants showed a decrease in the position dimension, and novice participants showed a decrement, particularly in the evidence dimension.

human-AI collaboration [5]. Accordingly, tools for supporting student researchers should automatically infer the perspective a user is currently emphasizing and provide targeted interventions that surface alternative viewpoints. This direction is particularly important for student researchers, who have typically been exposed to a narrower range of perspectives than senior researchers.

5 Limitations and Future Works

Given the small number of participants, our quantitative findings should be interpreted as indicative rather than definitive. Moreover, our current analysis relies primarily on focused reading time and interview reports and does not yet connect fine-grained LLM interaction behaviors to review quality. Future work should therefore recruit larger samples and analyze detailed LLM usage logs to characterize the mechanisms by which LLM interactions influence critical thinking and review quality. Additionally, our task operationalizes paper reading as preparation for review writing, whereas real-world research involves diverse reading goals (e.g., broad literature review) that demand different information and may elicit different uses of LLMs; thus, our findings may not generalize to settings that require synthesizing across multiple papers, motivating studies that focus on other objectives (e.g., literature review).

6 Conclusion

We conduct a user study to investigate how LLM assistance affects student researchers' critical thinking and how these effects vary with experience. We find divergent effects: for experienced student researchers, LLM assistance generally strengthens critical thinking, whereas for novices it does not consistently do so and may even be detrimental. These results point to the need for adaptive support that accounts for a student researcher's experience level. In summary, our study provides an initial step toward understanding how LLMs shape student researchers' growth.

References

- [1] Afza Diyana Abdullah, Xiaoting Qiu, Huan Li, and Muhammad Kamarul Kabilan. 2025. "It Becomes Very Intelligent": ChatGPT as an Academic Reading Tool for Postgraduates. *Reading Research Quarterly* 60, 4 (2025), e70058. arXiv:https://ila.onlinelibrary.wiley.com/doi/pdf/10.1002/rq.70058 doi:10.1002/rq.70058 e70058 RRQ-2024-11-0345.
- [2] Association of American Colleges and Universities. 2025. *How to Cite VALUE Resources*. <https://www.aacu.org/value/how-to-cite> Accessed: 2025-01-23.
- [3] Tara Capel and Margot Brereton. 2023. What is Human-Centered about Human-Centered AI? A Map of the Research Landscape. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 359, 23 pages. doi:10.1145/3544548.3580959
- [4] Celia Chen and Alex Leitch. 2024. LLMs as Academic Reading Companions: Extending HCI Through Synthetic Personae. arXiv:2403.19506 [cs.HC] <https://arxiv.org/abs/2403.19506>
- [5] Alexander Erlei, Abhinav Sharma, and Ujwal Gadiraju. 2024. Understanding Choice Independence and Error Types in Human-AI Collaboration. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 308, 19 pages. doi:10.1145/3613904.3641946
- [6] Andreas Fügner, Jörn Grahl, Alok Gupta, and Wolfgang Ketter. 2022. Cognitive Challenges in Human-Artificial Intelligence Collaboration: Investigating the Path Toward Productive Delegation. *Information Systems Research* 33, 2 (2022), 678–696. doi:10.1287/isre.2021.1079
- [7] Gaole He, Lucie Kuiper, and Ujwal Gadiraju. 2023. Knowing About Knowing: An Illusion of Human Competence Can Hinder Appropriate Reliance on AI Systems. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 113, 18 pages. doi:10.1145/3544548.3581025
- [8] Hyeon Jeon, Jeongin Park, Sungbok Shin, and Jinwook Seo. 2025. Stop Misusing t-SNE and UMAP for Visual Analytics. arXiv:2506.08725 [cs.HC] <https://arxiv.org/abs/2506.08725>
- [9] Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdoo Yun, Seongjin Shin, Sungdong Kim, James Thorne, and Minjoon Seo. 2024. Prometheus: Inducing Fine-grained Evaluation Capability in Language Models. arXiv:2310.08491 [cs.CL] <https://arxiv.org/abs/2310.08491>
- [10] Donald A. Norman. 2005. Human-centered design considered harmful. *Interactions* 12, 4 (July 2005), 14–19. doi:10.1145/1070960.1070976
- [11] Dejan Ravšelj et al. 2025. Higher education students' perceptions of ChatGPT: A global study of early reactions. *PLOS ONE* 20, 2 (02 2025), 1–53. doi:10.1371/journal.pone.0315011
- [12] Mark O. Riedl. 2019. Human-centered artificial intelligence and machine learning. *Human Behavior and Emerging Technologies* 1, 1 (2019), 33–36. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/hbe2.117 doi:10.1002/hbe2.117
- [13] Max Schemmer, Patrick Hemmer, Niklas Köhl, Carina Benz, and Gerhard Satzger. 2022. Should I Follow AI-based Advice? Measuring Appropriate Reliance in Human-AI Decision-Making. arXiv:2204.06916 [cs.HC] <https://arxiv.org/abs/2204.06916>
- [14] Ben Shneiderman. 2022. *Human-centered AI*. Oxford University Press.
- [15] Jiyao Wang, Haolong Hu, Zuyuan Wang, Song Yan, Youyu Sheng, and Dengbo He. 2024. Evaluating Large Language Models on Academic Literature Understanding and Review: An Empirical Study among Early-stage Scholars. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 12, 18 pages. doi:10.1145/3613904.3641917